

A
COMPREHENSIVE REPORT
ON
APPLICATION DEVELOPMENT AND EMERGING
TECHNOLOGIES

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BSCS-3B

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EXECUTIVE SUMMARY

This report analyzes and provides information about emerging technologies based on the Module 1 lecture to act as a guide for strategic planning. Emerging technologies are defined as a continuous advancement of technology and can be characterized by their novelty, fast growth, coherence, impact, and inherent uncertainty, as stated by Rotolo, Hicks, and Martin (2015). Key drivers of innovation such as AI, AR, and quantum computing alter how systems operate in the future. Generally, these innovations form the foundation of the fourth industrial revolution through big data analytics, the Internet of Things, cloud computing, and AR. Understanding the connection of these technologies is important because of how they continuously advance, and shape the global environment.

1. INTRODUCTION

Technologies continue to evolve rapidly, which has transformed people's way of living. Emerging technologies are defined as technologies that continue to advance each year. It is an innovation that is still in the process of developing, which have an impact on the environment, society, and economy in the upcoming years. Its definition can mean different things to other individuals. The purpose of this report is to provide a comprehensive overview of the emerging technologies from the Module 1 lecture, discussing their concepts, characteristics, areas, and industrial revolutions.

2. UNDERSTANDING EMERGING TECHNOLOGY

2.1 DEFINITION AND CONCEPT

According to the [businessdictionary.com](https://www.businessdictionary.com/definition/emerging-technology.html), emerging technologies are new technologies that are still being developed or will be developed in the future. However, this is inaccurate because new technologies are not the only thing considered emerging technologies. Artificial Intelligence is cited as an example of emerging technologies for the reason that it existed in the 1950s, but is still progressing to the present. Concluding, it is not always new technologies that are the basis for labeling whether it is considered emerging, but rather whether it is still advancing or progressing in how it is used and applied, even if it was invented a long time ago.

2.2 CHARACTERISTICS OF EMERGING TECHNOLOGY

The following characteristics of emerging technologies can be used as a basis to consider a technology as "emerging". It can be defined with the (5) five key characteristics by Rotolo, Hicks, and Martin (2015).

Table 1. Characteristics of Emerging Technologies	
Characteristic	Description
Radical Novelty	• Technology is being used in a new way or purpose. • Using it as a new solution to other existing problems. • Other approaches are used compared to old technologies with the same purpose. • For example, artificial intelligence is used in different ways and approaches.
Relatively Fast Growth	• Technology advances at a rapid pace compared to non-emerging technologies. • Can be seen in other aspects such as: actors, public and private funding, outputs/prototypes, products and services. • For example, many use artificial intelligence from different places.
Coherence	• Uniting previously independent technologies and research. • Connecting the ideas or logic to make innovation. • For example, cloud, AI, robotics, and data analytics have different purposes but united as one to achieve a new purpose that can be very beneficial for any field.
Prominent Impact	• It has a huge impact on different sectors such as the economy, society and industry. • It can influence a huge change to both small or big fields. • For example, artificial intelligence has a huge impact on different sectors such as business, education or healthcare.
Uncertainty and	• There is no assurance on the future outcomes of a

Table 1. Characteristics of Emerging Technologies	
Ambiguity	technology. • Ambiguity exists because we still lack knowledge about the possible outcomes, as its uses are still not fixed or interchangeable. • Different groups have different understandings of a certain technology. • For example, a lot of people bring new ideas about how artificial intelligence can be used, but there is uncertainty if it can be possible to develop.

3. AREAS OF EMERGING TECHNOLOGY

The following areas of emerging technology have a possibility of changing practices of industries in future generations, as stated by the Economic World Forum:

3.1 AUGMENTED REALITY (AR)

Augmented reality is a combination of graphics and reality. It acts as a simulation in which you are able to see digital assets in the real world. The following are examples of augmented reality applications:

- A game called Pokemon GO allows you to walk outside and catch digital Pokemons in the real world using your smartphone.
- A graphical furniture item that can be simulated in a real-world environment to see what it will look like before buying it.
- A car display guide that shows on a screen for drivers to park their cars or drive safely.

3.2 ARTIFICIAL INTELLIGENCE (AI)

Artificial Intelligence is a technology that imitates the way humans think and process information. This includes the process of learning and reasoning. An example of an AI application is the following:

- **Machine Learning:** An application of AI that allows a machine to learn from experience and patterns, similar to how humans do. It can be used in healthcare, finance, marketing

services, and transportation. For example, in healthcare, new medicines can be developed with the help of machine learning algorithms at a faster pace because of their ability to recognize patterns from data.

3.3 QUANTUM COMPUTERS

These are computers that are fast at performing calculations and solving problems more efficiently compared to a normal computer. Quantum computers use “qubits” as their basic unit of computation, like bits (0 or 1) from normal computers. However, Qubits are more advanced because qubits can be both 0 and 1 at the same time, which is called “quantum superposition”. IBM, Google, Rigetti Computing, and IonQ are among the companies that have demonstrated the largest quantum computers.

4. STAGES OF INDUSTRIAL REVOLUTION

Production is constantly changing because of technology. These stages are called the “Industrial Revolution”, and it is defined as a process of change that transforms manufacturing through the integration of physical and digital systems. The following are the (4) four stages of the Industrial Revolution:

Table 2. Four Stages of Industrial Revolution	
First Industrial Revolution (Industry 1.0)	• It began in the 18th century. • Introduces mechanical production such as hydroelectric and steam.
Second Industrial Revolution (Industry 2.0)	• It began in the 19th century. • Introduces use of electricity.
Third Industrial Revolution (Industry 3.0)	• Began in the first computer era. • Automation started to advance through the use of IT and electronics. • Strengthened further due to

Table 2. Four Stages of Industrial Revolution	
	internet access and better connectivity.
Fourth Industrial Revolution (Industry 4.0)	• Present and ongoing development of technology. • Introduces IoT, robotics, AR, VR and AI. • Reshapes different groups in the production process such as suppliers, producers and customers.

4.1 BUILDING BLOCKS OF FOURTH INDUSTRIAL REVOLUTION

- **BIG DATA ANALYTICS**

Big data analytics is a process of collecting and analyzing large amounts of data in order to find patterns and discover useful information. Companies use this to make faster decisions, especially in managing customer needs, since large data can now be processed almost instantaneously.

- **INTERNET OF THINGS (IoT)**

The Internet of Things is about tangible things that can be connected to the internet. It consists of sensors or wireless connections that allow devices to communicate with not only each other but also with a central system, making it possible to respond in real-time.

- **CLOUD COMPUTING**

It provides services online through an internet connection, such as storage or software, without the need to install it on your device. More data and processes are now being moved to the cloud, which allows production systems to rely on real-time, data-based services.

- **AUGMENTED REALITY (AR)**

It is used to help workers to do their jobs more efficiently. For example, guiding them to locate a specific item in a warehouse or giving repair instructions through a mobile device. It is still improving, and there is a possibility that companies will use AR in the future to give workers support for their decision-making at work.

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- Machine Learning is a type of artificial intelligence that enables computers to learn from data and patterns, similar to how humans learn from experience.

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